Lab: React Setup, Structure & Static Components

Instructor: Kashif Raza
Focus: Vite setup, folder architecture, JSX rendering, props, array mapping
Constraint: No hooks, no state. Pure functional components.

Step 1: Create & Clean the App

Open your terminal and run:

```
npm create vite@latest react-lab -- --template react
cd react-lab
npm install
npm run dev
```

Clean up: Delete all content inside `src/App.css` and `src/index.css`. Keep the files, just empty them.

Step 2: Target Folder Structure

Create this exact structure inside your `src/` folder:

```
src/
├── main.jsx
├── App.jsx
└── components/
    ├── TodoItem.jsx
    └── ProfileCard.jsx
```

Rule:

- `components/` holds reusable UI blocks.
- `App.jsx` holds layout, data arrays, and composition logic.
- `main.jsx` only mounts `App` to the DOM. Never put UI logic here.

Step 3: Bare Minimum Starter Code

Paste these into their respective files. This is your launchpad.

`src/main.jsx`

```jsx
import React from 'react'
import ReactDOM from 'react-dom/client'
import App from './App.jsx'
import './index.css'

ReactDOM.createRoot(document.getElementById('root')).render(
  <React.StrictMode>
    <App />
  </React.StrictMode>,
)
```

`src/App.jsx`

```
function App() {
  const todos = [
    { id: 1, task: "Set up Vite project" },
    { id: 2, task: "Create component folders" },
    { id: 3, task: "Render static lists" }
  ]

  const user = { name: "Kashif", role: "MERN Developer", status: "Active" }

  return (
    <div className="app">
      <h1>React Lab</h1>
      <section className="todo-section">
        <h2>Todo List</h2>
        <ul>
          {/* Task 1: Map todos here */}
        </ul>
      </section>

      <section className="profile-section">
        <h2>User Profile</h2>
        {/* Task 2: Render ProfileCard here */}
      </section>
    </div>
  )
}
export default App
```

`src/components/TodoItem.jsx`

```
function TodoItem({ item }) {
  return (
    <li>
      <input type="checkbox" />
      <span>{item.task}</span>
    </li>
  )
}
export default TodoItem
```

`src/components/ProfileCard.jsx`

```
function ProfileCard({ data }) {
  return (
    <div className="card">
      <h3>{data.name}</h3>
      <p>Role: {data.role}</p>
      <span className={`badge ${data.status.toLowerCase()}`}>{data.status}</span>
    </div>
  )
}
export default ProfileCard
```

Step 4: Your Tasks

Task 1: Build the Todo List

1. Import `TodoItem` at the top of `App.jsx`
2. Inside the `<ul>`, replace the comment with a `.map()` that returns `<TodoItem key={item.id} item={item} />`
3. Add this to `index.css`:

```
ul { list-style: none; padding: 0; }
li { display: flex; align-items: center; gap: 8px; padding: 8px 0; border-bottom: 1px solid #eee;
}
```

Task 2: Build the Profile Card

1. Import `ProfileCard` at the top of `App.jsx`
2. Inside the profile `<section>`, render `<ProfileCard data={user} />`
3. Add this to `index.css`:

```
.card { background: #f9f9f9; padding: 16px; border-radius: 8px; width: fit-content; }
.badge { padding: 4px 8px; border-radius: 4px; font-size: 12px; font-weight: bold; }
.badge.active { background: #d4edda; color: #155724; }
```

Step 5: Verification Checklist

- [ ] `npm run dev` shows no terminal or console errors
- [ ] `components/` folder contains exactly two files
- [ ] `App.jsx` imports both components
- [ ] `.map()` renders exactly 3 todo items
- [ ] Each `<TodoItem>` receives a unique `key` prop
- [ ] Profile card displays name, role, and status badge
- [ ] No `useState`, `useEffect`, or class components used

Deliverable: A clean, static React app demonstrating proper folder isolation, prop drilling, and array-to-JSX mapping. Push to GitHub and move to the hooks lecture.

Resources

- React Official Documentation
- Vite Documentation
- JSX in Depth
- Rendering Lists